

Repainting the Spider-Verse

A new painting FX pipeline with Rebelle and Houdini

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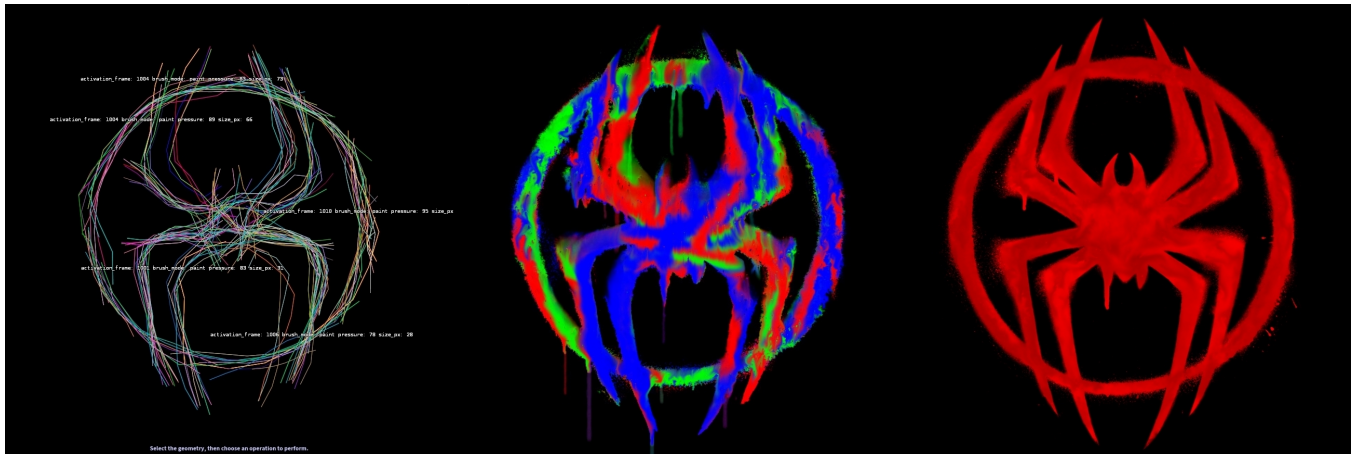


Figure 1: Curves with attributes applied in Houdini (left). Painting simulation in Rebelle (middle). Final image (right).

ABSTRACT

For *Spider-Man: Across The Spider-Verse*, we introduce a new method for creating a non-photorealistic look, simulating the movement of brushes and flow of paint. Instead of past approaches that occur in a later stage of a VFX pipeline, such as applying a shader or using a post-processing filter, our approach generates paint simulation data in the FX stage before lighting, integrating a modified version of Escape Motion’s Rebelle software with Houdini and OpenCue.

CCS CONCEPTS

• **Computing methodologies** → *Procedural animation; Physical simulation; Non-photorealistic rendering.*

KEYWORDS

Painting, Production, Pipeline, Simulation

ACM Reference Format:

Pav Grochola, Kenneth Nicol, Filippo Maccari, Peter Blaškovič, and Michal Fapšo. 2023. Repainting the Spider-Verse: A new painting FX pipeline with

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SIGGRAPH ’23 Talks, August 06–10, 2023, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-0143-6/23/08.

<https://doi.org/10.1145/3587421.3595451>

Rebelle and Houdini. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH ’23 Talks)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587421.3595451>

1 INTRODUCTION

To achieve a new look for *Spider-Man: Across The Spider-Verse* we took a new approach, thinking like traditional artists: instead of emulating a photographic outcome, emulate an artist’s process. Rebelle provided this process, utilizing the artist’s creativity to achieve an artistic outcome.

Rebelle [Blaškovič 2016] is a digital painting software that focuses on simulating traditional media, such as watercolors (Figure 2), oils, and acrylics. Its standout feature is its realistic watercolor simulation, which lets the user control the flow of paint and water on the canvas for blending, bleeding, and other watercolor effects. For oils, the software allows the adjustment of the amount of paint, oiliness, and impasto for thick or thin paint. It uses physical color mixing based on traditional pigments in a full RGB color gamut [Šárka Sochorová and Jamriška 2021], providing vivid colors that naturally produce secondary hues and avoid gray. Rebelle employs NanoPixel technology, an image recognition machine learning algorithm that enables up to 20x real-time macro zooming into the canvas and exporting 16x larger images.

While artists used Rebelle directly to paint and draft ideas, we also integrated it into Houdini to animate painting over time. Artists



Figure 2: Watercolor example. ©Karen Bonaker

could combine multiple tools in Houdini to get complex results, and paint in screen-space or on virtual canvases bound to world-space grids. Driving the simulation with a familiar node-based workflow resulted in an artist-friendly pipeline. Integrating Rebelle with the Sony Imageworks VFX pipeline allowed FX artists to bring more traditional artistic techniques together with modern tools to better express their vision for *Spider-Man: Across The Spider-Verse*.

2 DESIGN AND IMPLEMENTATION

2.1 Pipeline version 1

In our first attempt to integrate Rebelle with our VFX pipeline, we set up a Windows-Linux hybrid system based on prior work for the short film, *In Vaulted Halls Entombed*. Using OpenCue, a shell command to launch the Rebelle application was submitted from Houdini and executed on a render farm machine specifically built to run Windows and Rebelle. The command also ran a Python script to read a yaml file, generated from curves and attributes to define brush strokes, and use the pyautogui library to control the GUI widgets of Rebelle. This GUI control script was error-prone if the widgets were not in the expected configuration.

2.2 Rebelle batch mode

Next, a new mode in Rebelle batch-processes an external flow of events and commands created by nodes in a Houdini pipeline. An input json file contains commands for Rebelle, including pointer input events, changing paint tools, parameters, and brush color and others. The ability to import/export simulation data was also added, to let Rebelle and Houdini pass data to each other and work together to affect the next frame. Data types include RGBA and impasto of wet and dry layers, the amount of water and fluid velocity.

2.3 Pipeline version 2

In Houdini, artists author curves and apply point attributes, such as brush type and activation frame (Figure 1 left). A Houdini export process iterates over the points, maps the attribute values to a set of instructions, and writes the json file that Rebelle takes as input. A shell command to launch the Rebelle container is sent through OpenCue, to our render farm. Optionally, a VNC viewer connection is made to let the artists view the simulation's progress or debug problems. A limited palette of colours is used for the painting simulation (Figure 1 middle), but these are remapped later in the compositing stage with Nuke (Figure 1 right).

3 DISCUSSION

Rebelle was originally designed for real-time painting, not animation batch rendering like Houdini. So the main challenge was to translate these tools, very different in nature, into a productive workflow. It wasn't limited to migrating existing Rebelle features into the pipeline but also making additions required for VFX animation. Rebelle was highly optimized for painting in 8 bits per channel, but 16 bit channels were needed to improve results in compositing. Also, while the use of random generators gave a natural variability in the paintings, for VFX work we needed reproducible pseudorandom results.

Running containerized software via OpenCue for the first time in our pipeline provided challenges as well. Requirements for the new container system conflicted with the existing software stack, and additional container-related processes were spun off, requiring special treatment when canceling an OpenCue job.

There were many devils in the details to deal with as we iterated on the changes to Rebelle and our pipeline. Feedback from artists using the system has been positive. We wanted to see what would happen when we tried to emulate the artistic process of painting inside a VFX pipeline, and we found that artists appreciated the fast, creative workflow.

ACKNOWLEDGMENTS

Escape Motions: Thanks to our Dev team members: Miroslav Sedlák, Pavol Obuch, Anton Poživenec, Rastislav Vejo and Martin Surovček. Sony Pictures Imageworks: Thanks to our Production, Systems, Dev and PSR team members, including Steve Vargas, Sosh Mirsepassi, Alejandro Galindo, Tony Singh, Alexander Prigarin, Roula O'Regan, Noel Eaton.

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A ONLINE RESOURCES

Rebelle: <https://rebelle.escapemotions.com>